DOST Form 2 (for Basic/Applied Research)

DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

(1) PROJECT PROFILE

Project Title: **Impact Assessment of the Project on Development on Sustainable Breeder Philippine Mallard Duck (PMD) Production System**
Project Leader/Sex: **Roperto S. Deluna Jr/ Male**
Project Duration: 18 Months
Project Start Date: September 2023
Project End Date: March 2025
Implementing Agency: College of Applied Economics, University of Southeastern Philippines
Address/Telephone/Fax/Email: F. Inigo St. Davao City, Region XI

(2) COOPERATING AGENCY/IES (Name/s and Address/es)

Francis Ignatuis Nieva/Male
Partido State University/Goa, Camarines Sur, Region 5

(3) SITE(S) OF IMPLEMENTATION

IMPLEMENTATION SITES NO.	COUNTRY	REGION	PROVINCE	MUNICIPALITY
1.	PH	Central Luzon (R3)	Nueva Ecija	Zaragosa
2.	PH	Calabarzon (R4A)	Quezon	Candelaria
3.	PH	Calabarzon (R4A)	Laguna	Calamba
4.	PH	Caraga (R13)	Agusan del Sur	Trento
5.	PH	Davao Region (R11)	Davao de Oro	Compostela Valley
6.	PH	Calabarzon (R4A)	Quezon	Tiaong
7.	PH	Central Luzon (R3)	Nueva Ecija	Muñoz (CLSU)
8.	PH	Western Visayas (R6)	Iloilo	La Paz (WVSU)
9.	PH	Zamboanga Peninsula (R9)	Zamboanga del Sur	Zamboanga City (WMSU)
10.	PH	Davao Region (R11)	Davao del Sur	Davao City

(4) TYPE OF RESEARCH

_____ Basic
_✓__ Applied

(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)

__✓__ Agriculture, Aquatic and Natural Resources
Commodity: _____
_____ Health
Priority Topic: _____
_____ Industry, Energy and Emerging Technology
Sector: _____
_____ Disaster Risk Reduction and Climate Change Adaptation
_____ Basic Research
Sector: _____

Sustainable Development Goal (SDG) Addressed	No Poverty; Zero Hunger; Decent Work and Economic Growth; Industry, Innovation and Infrastructure; No Poverty; Sustainable Cities and Communities

(6) EXECUTIVE SUMMARY (not to exceed 200 words)

This impact assessment proposal aims to trace the utilization of the outputs and evaluate the impacts of the DOST PMD project implemented from 2012 to 2017. This will cover the five (5) implementation sites and five (5) DOST accredited suppliers across the three main islands of the Philippines. The main goal of the program is to improve the production capacities of the Philippine Mallard Duck Farms. Now, after 6 years from completion of the program, it is imperative to determine the impact of the investment on its beneficiaries and stakeholders. The impact assessment will be conducted to determine the achievement of the program's development objectives; assess the impact of the interventions on the beneficiaries at the micro, meso, and macro levels; and document lessons learned in the implementation of the program which are crucial input in designing similar or related interventions in the future. The total cost for the implementation of this impact study is PhP 6,410,078.50 (PhP 4,997,078.50 is requested from DOST-PCAARRD, while Php 706,500.00 and Php 706,500.00 are the counterparts of the University of Southeastern Philippines and Partido State University, respectively) to be implemented in 18 months across ten (10) study sites in the country.

(7) INTRODUCTION

In the Philippines, its duck population is composed of two genera namely the Mallard (*Anas platyrhynchos L.*) and the Muscovy (*Cairina Moschata L.*) and in the year 2009 the total population of ducks in the Philippines was at 10.37 million. The Philippine Mallard Duck also known as "Pateros" or "Itik" is a common type of duck that is being raised in the Philippines. And as the Philippine duck industry is dominated by balut (partially hatched embryos) production and by smallholder production that accounted for more than 75 percent of the total duck production. However, the commercial sector is gaining momentum in moving towards a higher degree of vertical integration and contract farming. At present, almost all the demand for duck eggs is met by domestic production. As such, imports appear not to be a serious threat to the domestic supply at present. However, it is envisaged that as trade liberalization continues, the Philippine duck industry will face increasing competition from overseas and other food products. Continuing survival and future growth of the industry depend on its ability to compete in a free trade environment, which, in turn, depends on efficiency of the production and marketing systems relative to its competitors (Dagaas et. al., 2004).

The Philippine duck industry is fragmented because of the large number of small-scale producers and traders involved in the production, processing, and marketing of duck products. Unlike their commercial chicken counterparts, the duck sector tends to rely on traditional and much less efficient production and marketing methods. Because more than three quarters of the ducks in the Philippines are raised in small-scale backyard production, there are no dominant players in the duck industry at the national or regional level. Continuing survival and future growth of the industry depend on its ability to compete in a free trade environment, which, in turn, depends on efficiency of the production and marketing systems relative to its competitors (Dagaas et. al., 2004).

In view of that, agricultural productivity and efficiency has been the center of many debates, policies and measures concerning the farming sector. The emphasis placed by the Sustainable

Development Goals on agricultural productivity underlines the many reasons for which additional research on statistical frameworks for productivity and efficiency targeted to developing countries is necessary. Thus, the DOST-PCAARRD has granted the BAI-NSPRDC with a project, entitled, “Development on Sustainable Breeder Philippine Mallard Duck (PMD) Production System”.

Through this project, the BAI-NSPRDC has developed a breeding true-to-type populations of layer-type Mallard duck leading to the development of Philippine signature egg-type duck breeds. With this breeding structure, it has improved the egg-laying performance of the Philippine Mallard ducks by 5% per generation and was able to establish a sustainable breeder duck and day-old duckling production and distribution system.

(7.1) RATIONALE/SIGNIFICANCE (not to exceed 300 words)

The Development of Sustainable Philippine Mallard Duck Production System is a 5-year DOST-PCAARRD funded project with a total budget of 10,767,654 *(PCAARRD only)* that ended in 2017. The project aims to offer sustainable solutions to the problem of poor and inconsistent production performance of the Philippine Mallard duck through genetic improvement and development of signature Philippine duck breed.

From 2015, the project has been able to improve the egg-laying performance of the Philippine Mallard ducks by 5% per generation thru an organized selection and breeding in the production of breeder ducks, develop breeding-true-to type populations of layer-type Mallard duck leading to the development of Philippine signature egg-type duck breeds, establish a sustainable breeder duck and day-old duckling production and distribution systems, and evaluate the profitability of quality breeder day-old-duckling and ready-to-lay pullet production enterprise.

After the project’s implementation and the amount of investment, an ex-post assessment is imperative. Thus, assessments of agricultural projects will evaluate and track how the allocated funds were utilized, what products were produced, who benefitted from these outputs, and how much benefit was derived by its beneficiaries. This information allows proper attribution of economic and societal gains due to the project.

These findings will be highly significant to DOST-PCAARRD in terms of future project disposition, funds allocation, and planning for future activities, hence enhance the effectiveness and efficiency of the program. Moreover, the determination of the program’s cost and benefits can be used to justify public investment of this nature (Piric and Reeve, undated) in the future.

(7.2) SCIENTIFIC BASIS/THEORETICAL FRAMEWORK

To measure if the project has achieved their expected outputs and outcomes, the conceptual framework to be used in the impact evaluation of the PMD project is focused on determining the interventions introduced or implemented by the project. In the case of the PMD project, its interventions of technology generation and transfer, capacity building, and practices *if adopted* will generate changes in quality, practices, and policies that will lead to the achievement of the overall program goal. The main goal of the program is to improve the production capacities of the Philippine Mallard Duck farms in the country. Eventually, the achievement of the goal will

produce micro (farm), meso (community level), and sectoral impacts that will result in improvement in quality of life.

The preceding outputs and outcomes of the project are premised on the assumption that no intervening factors will affect the program. These intervening factors may either positively or negatively affect the outputs and outcomes of the program. The inclusion of the intervening factors in the usual input-output-outcome scheme of impact assessment may be illustrated in Figure 1.

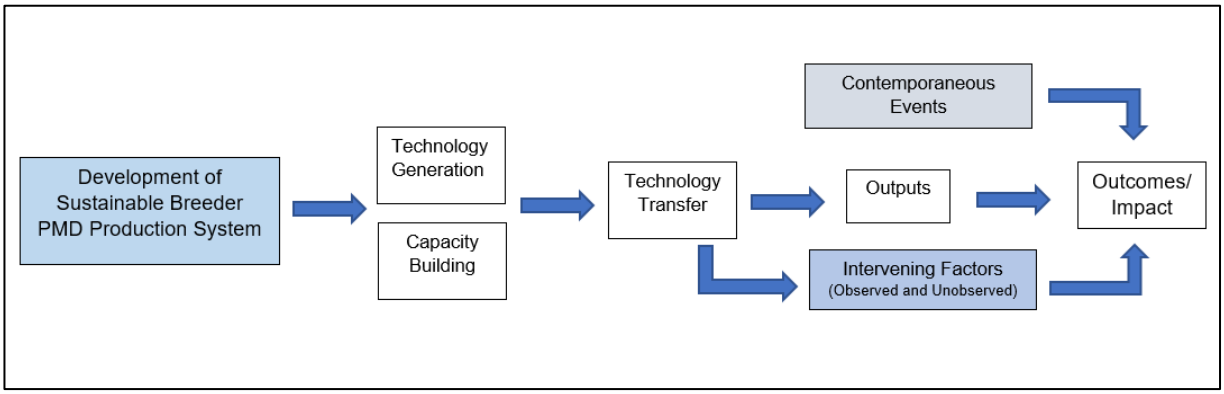


Figure 1. General conceptual framework of PMD project impact evaluation
Adapted from: Kene Ezemenari, Anders Rudqvist, K. Subbarao, "Impact Evaluation: A Note on Concepts and Methods", PRMPO, Poverty Reduction and Economic Management Network, The World Bank, January 1999.

(7.3) OBJECTIVES

General Objective:
The overall objective of the study is to assess the impact of the Development on Sustainable Breeder Philippine Mallard Duck (PMD) Production System Project.

Specifically, the study aims to:

1. Describe the processes and dynamics involved in the conceptualization, formulation, and implementation of the PMD project.
2. Describe the different component of the PMD project in terms of inputs, activities, and outputs.
3. Identify the techno-transfer modalities and track the status and/or subsequent use of the outputs produced.
4. Identify the adoption pathways (e.g., commercialization, capacity-building, communication, regulation) and track related projects and collaborators that facilitated the adoption of outputs.
5. Determine the economic (increase in the productivity), socio-institutional (e.g., social cohesions, value chain enhancement), and environmental impacts of the PMD project at the micro (i.e., farm), meso (i.e., consumer), and macro (i.e., industry) levels.
6. Documents lessons learned (hindering and facilitating factors); and
7. Provide policy recommendations that can improve the design and implementation of future R&D and technology transfer programs for the development of the PMD industry in the Philippines.

(8) REVIEW OF LITERATURE

This section will present a few reviews of related literature on the field of duck production, focusing on the impact of farming techniques, technology transfers, and breeding structure. However, there are limited research on the impact of breeding structure in duck production. Nevertheless, this section provides an overview of existing studies on the impact of farming techniques and technology transfers, while highlighting the gap in research on breeding structure. Thus, research is needed to enhance our understanding of this complex interplay.

In the overview study conducted at Cambodia focusing on the domestic duck production system concluded that there are only few farmers who uses improved breeds for upgrading their flocks; the majority rely on local birds, which they buy from nearby markets, neighbors or commercial farms. They prefer purchasing young ducks over older ones. The study also makes note of the nearly equal distribution of labor between men and women who manage and market the birds. And to determine if the duck populations in the five provinces are comparable or dissimilar, the study used multivariate analysis. To calculate the Mahala Nobis distances between provinces based on all phenotypic traits, a canonical discriminant analysis was used. Additionally, a stepwise discriminant analysis reveals which variables are most important for separating provinces. To distinguish the various production systems and describe the ducks in each, the acquired data were individually analyzed for each province. Four categories of information are used to describe different production systems: farmer households and lifestyle, livestock totals, management, and marketing. (Dinesh, et al, 2009).

To assess the effectiveness of the technology transfer in this case the vocational training program on integrated fish farming, Singh and Tanwar also conducted an international impact evaluation on integrated fish-cum-duck farming. A knowledge exam made up of ten objective-type questionnaires was developed to assess the effectiveness of training programs before and after the training. Marks were assigned quantitatively for each response provided by the respondents. The knowledge score for each participant was determined by giving one credit for a correct answer and none for an incorrect answer. So, ten was the highest possible score on the test. Furthermore, the impact evaluation's findings show that Krishi Vigyan Kendra, Barnala's vocational training programs were effective in spreading contemporary sustainable agricultural methods at the local level. Most participants in these training programs were farmers and workers, who can use this expertise to establish self-employment and close the productivity and production gap, supporting the nation's economy even more (Singh & Tanwar, 2020).

Similar to this work, in 2008, Lavega, et al. used a bio-economic simulation model to assess several management strategies for smallholder mallard duck production systems in the province of Iloilo. Egg production grew as the number of duck layers increased, according to the economic implications of the simulation for the three management methods. Gross sales were calculated for each management system simulation using the reported average price of an egg per piece (PhP3.20). Gross sales in the baseline simulation were PhP16,198, which were 30% lower than gross sales because of management intervention I (PhP21,062.40) and 49% lower than gross sales because of management intervention II (PhP17,048.80). (P24,118.40). When gross sales of both interventions were compared, a difference of about 14% was observed. Additionally, the mortality rates in both the brooding and developing stages of the ducklings were reduced by complete confinement and semi-confinement, respectively. Egg production increased in response to an increase in the number of laying hens. Therefore, it is determined that confinement is the optimum alternative management technique for enhancing duck smallholders' production system. In Iloilo, smallholder duck farming uses a traditional method; contemporary technology is not used. Assessing management choices to increase duck production, which could subsequently assist improve the socioeconomic position of the rural community, is greatly aided by modeling and simulation. Additionally, Iloilo's Small hold Duck Production uses a

traditional system and does not use any modern technologies. Assessing management choices to increase duck production, which could subsequently assist improve the socioeconomic position of the rural community, is greatly aided by modeling and simulation.

In addition, a study was done to evaluate Itik PINAS (IP)-Kayumanggi's laying performance, egg quality, fertility rate, and egg classification under the intense and semi-intensive farming system. The collected data, which included information on production efficiency, egg quality, and hatchability, was examined using Statistica's T-Test with a level of significance set at P 0.05. In terms of egg classification, livability rate, and egg quality components, evaluation results preferred a semi-intensive production strategy. The results show that the newly created IP-Kayumanggi has a semi-intensive system performance benefit. Additionally, the production of balut will benefit from the increased fertility rate of eggs laid by IP-Kayumanggi ducks in a semi-intensive system. (Julian, et al.,2022). Furthermore, a study of duck egg production in the Philippines that used a descriptive research design to evaluate the factors influencing the decision-makers choices of stock replacement in duck egg production reveals that variations in regional preferences also led to variations in the raisers' methods for raising ducks. Most individuals who raised poultry or kept them in pens tended to use ready-to-lay pullets, whereas those who employed the free-range farming approach either used ready-to-lay ducks or day-old ducklings. (Gutierrez, 2019).

However, studies assessing the impact of PMD project is very limited. There has been anecdotal evidence of the positive outcomes that the project created but as to quantifying the impacts, the literature is very limited. Hence this proposed impact assessment study be a first step in evaluating its impact. It will map out the inputs and outputs of the projects and trace its intended and unintended outcomes and measure its associated impact. This made it true that the Philippine Mallard Duck industry has a large potential for growth that has yet to be realized.

(9) METHODOLOGY

Overall Evaluation Approach

To engage with the various PMD project partners and stakeholders, the approach on this evaluation is multi-sectoral, consultative, participatory, and stakeholder driven. The insights and opinions of the government, private sectors, itik farmers, and consumers, on the level of performance of the project’s implementation are crucial in responding to the multi-faceted key impact evaluation questions. The availability of project documents is crucial to the identification of study participants.

The Itik Pinas project's impact evaluation will be conducted using a mixed-methods approach. Using mixed methodologies, where quantitative and qualitative data play complementary roles, impact evaluation designs are reinforced. (White and Raitzer, 2017). A farm-level survey will be used to gather the quantitative data, as will the acquisition of secondary data from pertinent organizations. On the other hand, key informant interviews (KIIs), focus groups/small group discussions (FGDs), and the examination of project papers will produce qualitative data. Quantitative data will be analyzed primarily using Microsoft excel for cost and return analysis, stochastic frontier (SFA) for the technical efficiency analysis, Stata V15 for the Logistic regression and time series analysis and Modex-C for welfare analysis.

The evaluation questions can be answered by combining the findings of the qualitative and quantitative analyses adapting the MMIAI of RD&E interventions framework.

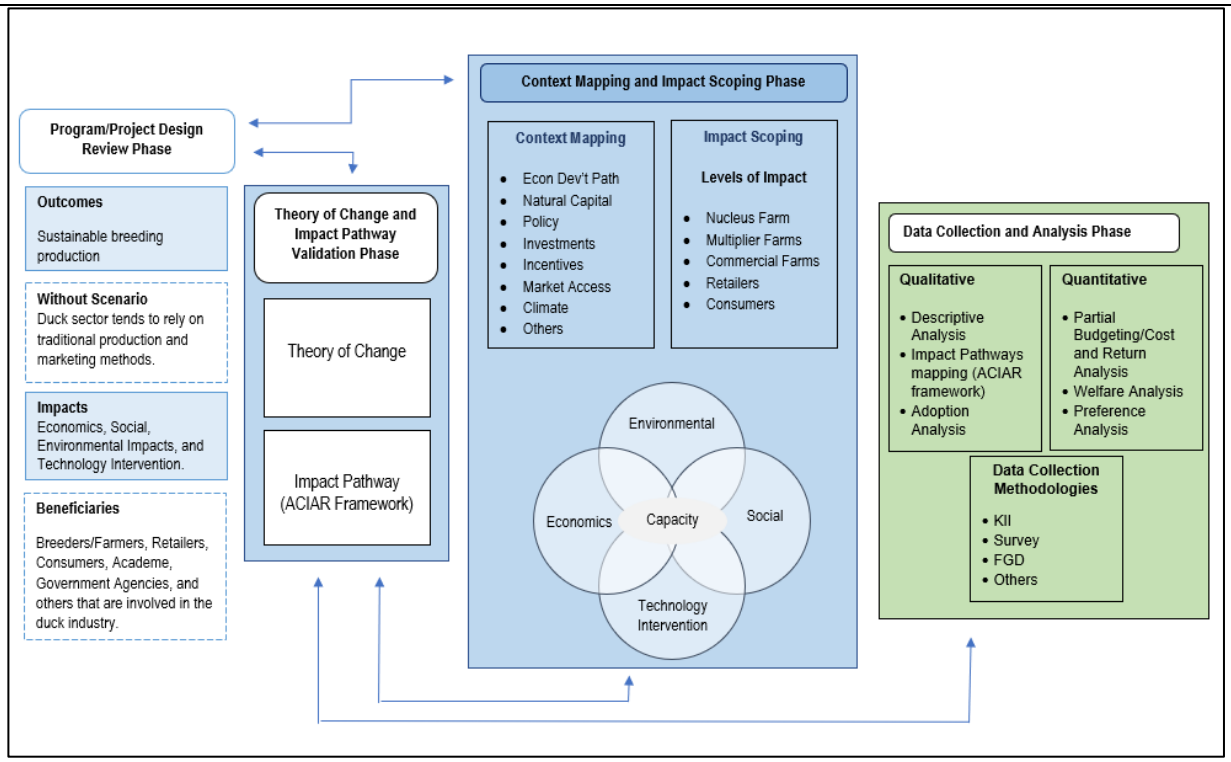


Figure 1. Overall evaluation approach

Adapted from: Williams L.J., et. al. (2021). An integrated approach to ex-post impact assessment. *ACIAR Impact Assessment Series Report No. 102*

The main framework to be used in assessing the impact of PMD project is anchored on the framework developed by Davis et al. (2018) and used by the Australian Centre for International Agricultural Research (ACIAR). This framework outlines the description of how a project is supposed to deliver the desired results. It reflects the theory of change by describing and mapping how and why a particular project, modality or design will attain its intended (or unintended) outcomes. As one of the major steps in impact evaluation, mapping project inputs to outcomes is necessary before quantifying the impacts.

This approach details the causation link between the inputs of the intervention up to its net benefits (Figure 2). Starting from the inputs, the resources or investments that have been placed for the intervention are analyzed. They are valuated in currency forms to use a universal unit. Next to the input analysis is the output which investigates the actual product of the intervention. This may be a technology, a policy, or a capacity. Outputs are then adopted through different adoption pathways that can be commercialization, communication, capacity building, or regulation. Once adopted, outcomes will be analyzed. Outcomes can be change in product, policy, or practice. From these outcomes, impacts can be derived and determined. And from these determined impacts, the net benefits or loss can be identified.

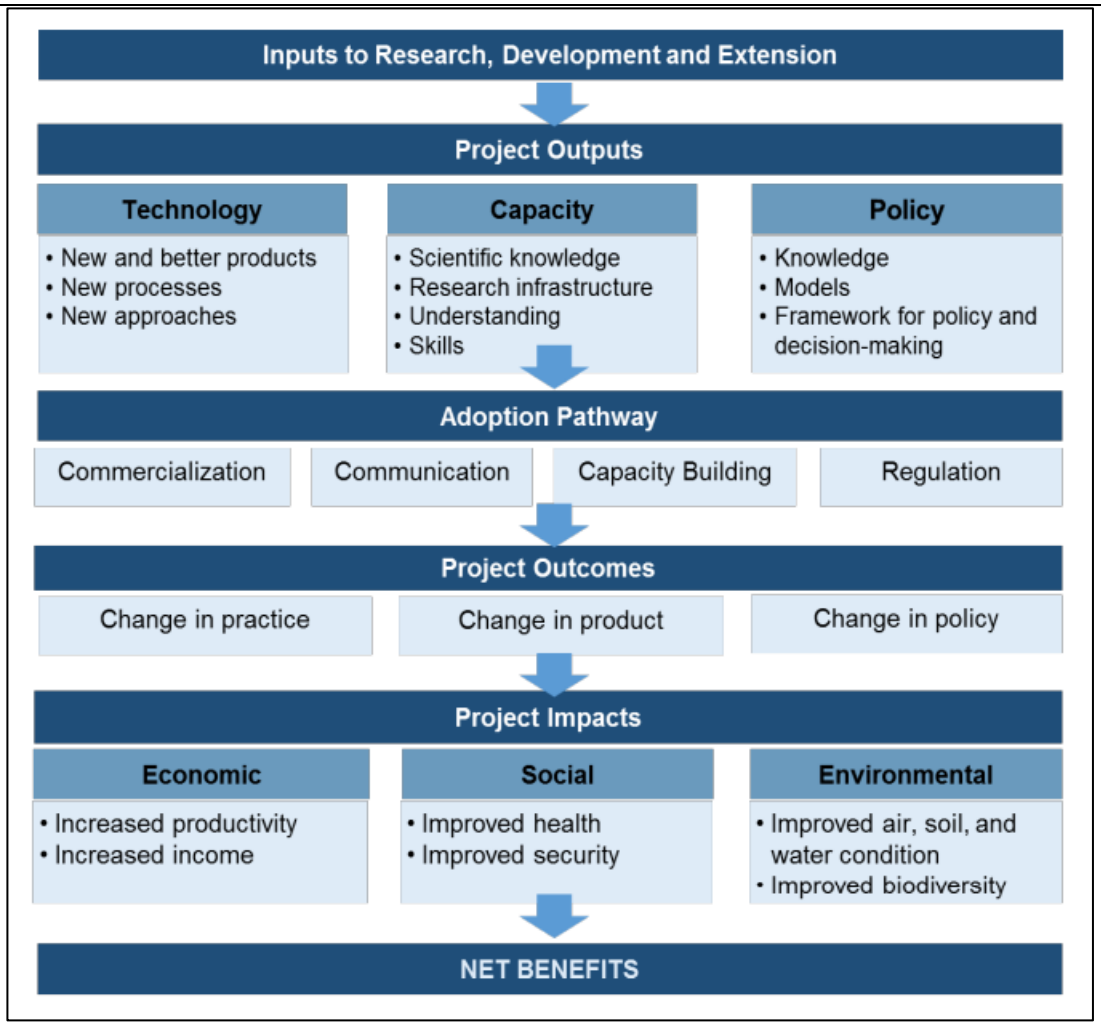


Figure 2. Framework for Impact Assessment (Davis et al., 2008)

Sources of Data and Methods of Data Collection

Both primary and secondary data will be utilized in the study. Primary data will be obtained from stakeholders' surveys using interview schedules programmed in the Kobo toolbox designed for the different types of respondents. Initially, based on the available documents and the nature of interventions provided by the project, there are four (4) types of survey respondents from different components of the PMD value chain:

a. Nucleus Farm,
b. Multiplier Farm,
c. Commercial Farm,
d. Retailers,
e. Consumers.

All implementation sites will be covered by the surveys. The details on the sample size per type of survey will be determined once program documents particularly the program terminal reports are available to the evaluation
team. Pre-testing of the questionnaires will be conducted before the actual survey/ field activities.

Basic data to be generated, but not limited to, include the following:

a. socio-demographic profile of the respondent and the immediate household members (e.g., name, age, gender, civil status, household size, number of children, number of children attending/not attending school, educational attainment, and occupation);

b. farm ownership, tenurial status; farm size; farm location, membership in the organization; name and type of organization; forms of assistance provided under the PMD project,
c. the training attended under the project; technical assistance received (improved practices and PMD breeding protocols), and other inputs, and marketing assistance received;
d. household income by sources (day-old egg/itik income, other agricultural sources of income, and non-farm);
e. technologies introduced under the PMD project which were adopted productivity, the price received, gross income and cost of production by type of agricultural enterprise and by technology adopted;
f. access to infrastructure facilities to support PMD breeding, and marketing;
g. household farm assets and personal assets including house condition and ownership, appliances, and jewelry; transportation units, farm animals, implements, and equipment. In the impact evaluation survey, the respondents will be asked to indicate which of the household farm and personal assets were partly or fully financed from the increased income derived from the project;
h. household expenditures focusing on those expenditures which were reduced due to program interventions;
i. employment generated, marketing channels, marketing structure, profit, and
j. all other indicators and measurements required in the empirical analysis of this impact study.

Secondary information will also be gathered to be used in the impact analysis. Visits to the different offices of the cooperating agencies will be made to collect relevant documents. These include the network of implementers of the PMD breeding industry strategic S&T plan composed of the following agencies:

a. Bureau of Animal Industry–National Swine and Poultry Research and Development Center (BAI-NSPRDC)
b. Central Luzon State University (CLSU)
c. West Visayas State University (WVSU)
d. Western Mindanao State University (WMSU)
e. Department of Agriculture Regional Field Office 11 (DARFO11)

Key informant interviews will include implementing agencies partners, and PMD focal persons of the agencies included in the PMD breeding industry strategic S&T plan. Focus group discussion will be done at the community level (local officials, farmers' groups, women's groups, and youth).

The use of focus group discussions as a tool will also be utilized during the consultative interviews that would be conducted at the various stages of the supply chain. The structure of the design of the FGD would be developed to encourage participatory discussions using tools to make the participants highly energized, productive, and engaged in meaningful participation throughout the workshop proceedings. Moreover, the following general agenda items will guide this participatory process:

a. Presentation of the initial findings expressed as facts and data derived from the review of related project documents and the conduct of the key informant interviews;
b. Elicitation of the initial reactions to the initial findings of the participants;
c. Elicitation of critical thinking on the result of the overall development impact and its concomitant meaning, value, and significance to the project; and,
d. Requiring and/or challenging the participants to provide for and decide for their individual and collective opinions and resolutions; and,
e. Proposing possible next steps, i.e., what can/will be done?

Site Visit/ Review of the Program Documents, Reports, Others

Since the study has been terminated a 5 years ago, there is a need for the members of the Assessment Team to be familiar with the project area and to have an ocular inspection of the area, and to look for themselves the perceived changes in the area.

To gather the necessary documentation, visits will be conducted to the various offices of the cooperating agencies and other service providers. Documents, including other relevant studies and evaluations, will be made available, with a focus on those relating to project formulation and implementation plans, annual work plans and budgets, manuals, log frames, baseline studies, and monitoring and evaluation reports. Various periodic status and accomplishment narrative reports, adjustments to implementation plans and budgets, policy issuances, partnership agreements, and other reports regarded to be similarly essential will also be covered during this section of the review phase.

Second, a multi-level review and analysis procedure that included the following steps would be used, namely:

a. On actual resources deployed, that is, organizational structures and functions; fund utilizations vis-à-vis approved budgets; monitoring, evaluation, and adjustment system; management information system; partnership agreements at all levels; contracts with resource institutions and individual consultants; trainings and other capacity-building exercises conducted; and, other mechanisms utilized;
b. On activities undertaken and delivery status, that is, actual versus planned activities per approved work and financial plans including exceptions such as slippages and unplanned activities;
c. On actual results through actual activities undertaken vis-à-vis expected results mainly in terms of benefits obtained on the transfer of institutional know-how through restructuring, the introduction of modern management techniques, and capacity-building activities, etc;
d. Identification of impact indicators or measures such as: the qualitative and quantitative levels of resource allocation and sharing and 'know-how' transfers to the project implementing partners and counterparts at all levels; their ability to absorb and benefit from these transfers that could ensure the sustainability of similar interventions in the future even without program assistance; and, the extent of the impact on the ultimate beneficiaries.

Lastly, documents related to similar foreign-assisted projects and/or other donor projects operating in the area would also be examined. These include their respective program formulation and implementation plans, annual work plan and budget; manuals; logical frameworks, etc.

The results of the KIIs, FGDs, and document reviews would provide sufficient quantitative data for the analysis.

Impact Pathway Mapping

The ACIAR guidelines developed by Davis et al. (2008) will be utilized in the sequential mapping of program inputs, outputs, outcomes, and impacts. This framework outlines the description of how a project is supposed to deliver the desired results. It reflects the theory of change by describing and mapping how and why a particular project, modality, or design will attain its intended (or unintended) outcomes. As one of the major steps in impact evaluation, mapping project inputs to outcomes is necessary before quantifying the impacts. One of the overarching principles for impact assessment is the clear identification of causal between levels of outputs,

the framework developed by Davis et al. (2008) clearly shows the causation link between the inputs of the intervention up to its net benefits.

Empirical Analysis

The empirical analysis will be utilized various estimation methodology to answer the objectives of this study. Presentation of the survey data and secondary data will utilize descriptive statistics (measures of central tendencies and measures of dispersion, ratios, and percentages, and growth rates) supplemented by graphics. This also includes adoption variables such as lag of adoption, the proportion of farmer adopters, rate of adoption, and related information. In addition, the study will use the following quantitative techniques, when appropriate:

a. t-test for paired samples will be used to determine the effects of the PMD p-project on the "before and after" situation on the factors related to the main components of the project;
b. Logit model will also be explored to determine the probability of awareness and attitude in the adoption of certain practices/protocols, PMD, and technologies introduced in the project;
c. Production function analysis to measure productivity and technical efficiency of duck farms with and without program interventions.
d. Partial Budget Analysis to identify the change in net income that is due to changes in cost and return items that were changed because of the adoption of project interventions. This analysis will be carried out at the PMD breeding production, and processing stages.
e. Total Factor Productivity (TFP) Analysis will be utilized to estimate TFP growth which measures the change in total revenue which cannot be explained by changes in output price and input used. The computation for the growth in TFP will be estimated in two (2) reckoning periods: the growth in TFP without versus during the PMD project, and during PMD project vs. current practice (now).
f. Consumer, producer, and total surplus analysis using the ACIAR's MODEXC program will be used to estimate the change in welfare resulting from a change in demand.
g. Benefit-Cost Analysis (BCA) will be used to analyze the monetary values of program benefits and the program investment to assess the project's financial viability using the internal rate of return (IRR), the net present value (NPV), and benefit-cost ratio (BCR) as measures of project worth.

Environmental Assessment

The study will look at the environmental effects brought on or prevented by the project interventions in addition to the economic effects of the program. The adverse environmental impacts of implementing a new duck farming breeding method may include increased pollution from waste production, habitat destruction from land conversion, genetic pollution from genetic modification or hybridization, increased resource use, such as water and energy, potential chemical contamination from antibiotic or pesticide use, higher energy consumption from heating or cooling systems, and improper waste disposal leading to pollution. To reduce these potential negative consequences and guarantee sustainable duck farming operations, it is essential to carry out extensive environmental evaluations and put appropriate management techniques into place. Threats to the environment only appear after decades; thus, it is essential to analyze previous environmental assessments conducted at the implementation sites both inside and outside of the program. This will be compared to the research's findings, which will be produced.

10) TECHNOLOGY ROADMAP (if applicable) (use the attached sheet)
Not Applicable

(11) EXPECTED OUTPUTS (6Ps)

Publication: At least one (1) draft journal article

Patent: No patent/IP output for this project

Product: Information on the program's impact on stakeholders particularly duck farmers, women, and youth

People Services: Ten (10) trained researchers.

Places and Partnerships: Will cover the 10 implementation sites (Laguna, Agusan del Sur, Davao de Oro, Quezon, Nueva Ecija, Iloilo City, Zamboanga City, and Davao City) for visits and potential collaboration to see the effect of localized context in the impact pathways.

Policy: Identification of areas that requires further policy intervention, policy recommendations on sustainability (e.g. institutionalization of the program introduced practices that are proven to be effective), policy recommendation to complement and support changes in outcomes caused by the program.

(12) POTENTIAL OUTCOMES

Improved and up-to-date design of PMD project and technology transfer programs.

(13) POTENTIAL IMPACTS (2Is)

Increase production capacities of duck farms in the country and increase in the contribution of the sector to the overall economic development of the country.

(14) TARGET BENEFICIARIES

a. Policy and decision-makers, national R&D system, and R&D funding agencies
b. Researchers and development workers
c. DOST-PCAARRD
d. Evaluators of R&D programs
e. Academe
f. Duck farming networks across the country

(15) SUSTAINABILITY PLAN

Not Applicable

(16) GENDER AND DEVELOPMENT (GAD) SCORE (refer to the attached GAD checklist)

The impact evaluation study will also examine the effect of the PMD project on the welfare of women and youth in the communities of the implementation sites. Instruments to be utilized in data gathering will be checked for gender sensitivity.

(17) LIMITATIONS OF THE PROJECT

Not Applicable

(18) LIST OF RISKS AND ASSUMPTIONS RISK MANAGEMENT PLAN (List possible risks and assumptions in attaining target outputs or objectives.)

Not Applicable

(19) LITERATURE CITED

Davis, J., Gordon, J., Pearce, D., & Templeton, D. (2008). *Guidelines for assessing the impacts of ACIAR's research activities*. ACIAR Impact Assessment Series Report No. 58, 120 pp.

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Position	Percent Time Devoted to the Project	Responsibilities
Program Leader	30%	Provide overall supervision of the evaluation team and ensure timely submission of deliverables to the funding agency.
Project Staff L3	30%	Assist in handling the conduct of impact study and executing the tasks and producing deliverables outlined in the project plan and schedule.
Project Staff L2	30%	Assist in project implementation.
S and T Consultants	15%	To provide technical inputs to the project particularly on the analysis of technical research outputs and interventions of the program.
Science Research Analyst	100%	To assist the project leaders and study leaders in technical matters relevant to the conduct of the study, analysis, and report writing.
Science Research Assistant	100%	To assist the team on technical and administrative requirements, to ensure timely submission of financial audited reports as required by the funding agency.

(20) PERSONNEL REQUIREMENT

(21) BUDGET BY IMPLEMENTING AGENCY

IMPLEMENTING AGENCY		PS	MOOE	EO	Total
a. DOST-PCAARRD	Year 1	1,552,920.00	1,915,116.00	65,000.00	3,533,036.00
	Year 2	776,460.00	687,582.50		146,4042.50
b. USeP	Year 1	186,000.00	285,000.00		471,000.00
	Year 2	93,000.00	142,500.00		235,500.00
c. ParSU	Year 1	186,000.00	285,000.00		471,000.00
	Year 2	93,000.00	142,500.00		235,500.00
					6,410,078.50

23) OTHER SUPPORTING DOCUMENTS WORK AND FINANCIAL PLAN

Tables 1 and 2 outlines the project activities and deliverables as well as the cash financial plan. The impact assessment will be conducted for 1.5 years (18 months).

Table 3. Work Plan (please see attached Workplan)

Table 4. Proposed Financial Plan

DOST Form 4

DEPARTMENT OF SCIENCE AND TECHNOLOGY
Project Line-Item Budget
CY 2022

Project Title : Impact Assessment of the Project on Development on Sustainable Breeder Philippine Mallard Duck (PMD) Production System
Implementing Agency : University of Southeastern Philippines
Total Duration : 18 Months
Current Durtion : May 2023 - November 2024
Project Leader : Roperto S. Deluna Jr (University of Southeastern Philippines)
Collaborating Agency : Partido State University
Funding Agency : DOST

	DOST							COUNTERPART						GRAND TOTAL		
	University of Southeastern Philippines			Partido State University			Total	USeP Counterpart			ParSU Counterpart					
	Year 1	Year 2	Total	Year 1	Year 2	Total		Year 1	Year 2	Total	Year 1	Year 2	Total	Year 1	Year 2	Total
I. Personal Services	P		P													
Direct Cost																
Salaries																
One (1) Science Research Analyst @P32,400/month	388,800.00	194,400.00	583,200.00			-	583,200.00			-			-	388,800.00	194,400.00	583,200.00
Two (2) Science Research Assistant @P25,355/month	304,260.00	152,130.00	456,390.00	304,260.00	152,130.00	456,390.00	912,780.00							608,520.00	304,260.00	912,780.00
Honorarie										-			-			
One (1) Project Leader @P8,800/month	105,600.00	52,800.00	158,400.00				158,400.00							105,600.00	52,800.00	158,400.00
Five (5) Project Staff Level 3 @P7,500/month	180,000.00	90,000.00	270,000.00	270,000.00	135,000.00	405,000.00	675,000.00	90,000.00	45,000.00	135,000.00	90,000.00	45,000.00	135,000.00	630,000.00	315,000.00	945,000.00
Administrative Support Staff								96,000.00	48,000.00	144,000.00	96,000.00	48,000.00	144,000.00	192,000.00	96,000.00	288,000.00
Sub-total for PS	P 978,660.00	P 489,330.00	P 1,467,990.00	P 574,260.00	P 287,130.00	861,390.00	2,329,380.00	186,000.00	P 93,000.00	279,000.00	186,000.00	P 93,000.00	279,000.00	1,924,920.00	P 962,460.00	2,887,380.00
II. Maintenance and Other Operating Expenses						-										
Direct Cost	**958,720.00**	**360,680.00**	**1,319,400.00**	**714,440.00**	**224,760.00**	**939,200.00**	**2,258,600.00**									
Traveling Expenses (Local)	306,600.00	131,400.00	438,000.00	151,200.00	64,800.00	216,000.00	654,000.00							457,800.00	196,200.00	654,000.00
Communication Expenses	30,000.00	15,000.00	45,000.00	24,000.00	12,000.00	36,000.00	81,000.00							54,000.00	27,000.00	81,000.00
Supplies and Materials Expenses	36,000.00	18,000.00	54,000.00	36,000.00	18,000.00	54,000.00	108,000.00	45,000.00	22,500.00	67,500.00	45,000.00	22,500.00	67,500.00	162,000.00	81,000.00	243,000.00
Utility Expenses (e.g., light, water, internet connection, etc.)								60,000.00	30,000.00	90,000.00	60,000.00	30,000.00	90,000.00	120,000.00	60,000.00	180,000.00
Rent Expenses (e.g., vehicles, venue, etc.)	112,000.00	48,000.00	160,000.00	105,000.00	45,000.00	150,000.00	310,000.00							217,000.00	93,000.00	310,000.00
Representation Expenses (e.g. food for meetings, etc.)	61,320.00	26,280.00	87,600.00	30,240.00	12,960.00	43,200.00	130,800.00							91,560.00	39,240.00	130,800.00
Survey Expenses	268,800.00		268,800.00	224,000.00		224,000.00	492,800.00							492,800.00	0.00	492,800.00
Professional Services	144,000.00	122,000.00	266,000.00	144,000.00	72,000.00	216,000.00	482,000.00							288,000.00	194,000.00	482,000.00
Other Maintenance and Operating Expenses						-										
Office Space						-		180,000.00	90,000.00	270,000.00	180,000.00	90,000.00	270,000.00	360,000.00	180,000.00	540,000.00
Indirect Cost (7.5% of PS +MOOE Direct Cost)	**145,303.50**	**63,750.75**	**209,054.25** #	**96,652.50**	**38,391.75**	**135,044.25**	**344,098.50**									
Traveling Expenses	101,712.45	44,625.53	146,337.98	67,656.75	26,874.23	94,530.98	240,868.95							169,369.20	71,499.75	240,868.95
Supplies and Materials Expenses	29,060.70	12,750.15	41,810.85	19,330.50	7,678.35	27,008.85	68,819.70							48,391.20	20,428.50	68,819.70
Communication Expenses	14,530.35	6,375.08	20,905.43	9,665.25	3,839.18	13,504.43	34,409.85							24,195.60	10,214.25	34,409.85
Sub-Total for MOOE	P 1,104,023.50	P 424,430.75	P 1,528,454.25	P 811,092.50	P 263,151.75	1,074,244.25	2,602,698.50	285,000.00	P 142,500.00	427,500.00	285,000.00	P 142,500.00	427,500.00	2,485,116.00	P 972,582.50	3,457,698.50
III. Equipment Outlay																
Computers	65,000.00		65,000.00				65,000.00							65,000.00		65,000.00
Photocopier																
Software																
Sub-Total for EO	P 65,000.00	P	P 65,000.00	P	P		65,000.00	P		P			P	65,000.00	P	65,000.00
GRAND TOTAL	P 2,147,683.50	P 913,760.75	P 3,061,444.25	P 1,385,352.50	P 550,281.75	1,935,634.25	4,997,078.50	471,000.00	P 235,500.00	706,500.00	471,000.00	P 235,500.00	706,500.00	4,475,036.00	P 1,935,042.50	6,410,078.50

I hereby certify the truth of the foregoing and have no pending financial and/or technical obligations from the DOST and its attached Agencies. I further certify that the programs/projects being handled is within the prescribed number as stipulated in the DOST-GIA Guidelines. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.

	SUBMITTED BY (Program Leader)	**ENDORSED BY (Head of the Agency)**
Signature		
Printed Name	**ROPERTO S. DELUNA, JR., PhD**	**LOURDES C. GENERALAO, PhD**
Designation/Title	**Assistant Professor**	**SUC President IV**
Date	June 1, 2023	